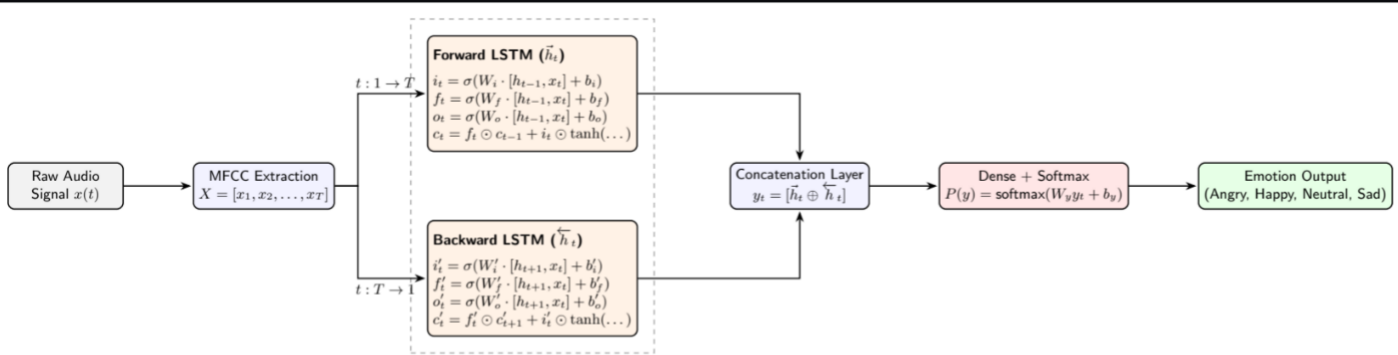

Raw Audio Signal $x(t)$
MFCC Extraction $X = [x_1, x_2, \ldots, x_T]$
$t : 1 \to T$
Forward LSTM ($\overrightarrow{h}_t$)
$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$
$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$
$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o)$
$c_t = f_t \odot c_{t-1} + i_t \odot \tanh(\ldots)$
$t : T \to 1$
Backward LSTM ($\overleftarrow{h}_t$)
$i'_t = \sigma(W'_i \cdot [h_{t+1}, x_t] + b'_i)$
$f'_t = \sigma(W'_f \cdot [h_{t+1}, x_t] + b'_f)$
$o'_t = \sigma(W'_o \cdot [h_{t+1}, x_t] + b'_o)$
$c'_t = f'_t \odot c'_{t+1} + i'_t \odot \tanh(\ldots)$
Concatenation Layer
$y_t = [\overrightarrow{h}_t \oplus \overleftarrow{h}_t]$
Dense + Softmax
$P(y) = \text{softmax}(W_y y_t + b_y)$
Emotion Output
(Angry, Happy, Neutral, Sad)